IFN647 ASSIGNMENT 2

FINAL REPORT

Group Number: 43

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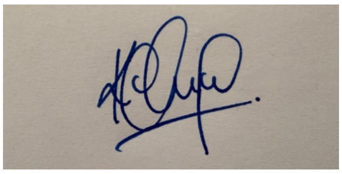
Statement of completeness and team agreement

Statement of Completeness Group Number:		
The undersigned group members agree to adhere to this statement to ensure the successful completion of IFN647 Assignment 2 in accordance with the project requirements and timelines. We confirm that each team member has made an equal or near-equal contribution to the project.		
Name, student number & email	**Signature**	**Date**
Janardhanan Yoganandan, N11918411, n11918411@qut.edu.au		*21/05/2026*
Chitesh Kolakaleti, n11907304, n11907304@qut.edu.au		*21/05/2026*
Task Allocation for each student *Janardhanan Yoganandan*	**Tasks and percentage of contribution (%)** 50% contribution - Set up the project folder structure and task files. - Implemented and checked Baseline1 BM25 model. - Implemented and checked Baseline2 JM language model. - Implemented and checked Model_C pseudo-relevance hybrid model. - Generated ModelOutputs ranking files for all topics. - Ran evaluation and significance testing. - Prepared and reviewed the final report sections and appendices.	
Chitesh Kolakaleti	50% contribution - Assisted with code testing and output checking. - Checked data folder setup, topic-dataset matching and generated ranking outputs. - Assisted with evaluation result checking, top-10 appendix preparation and validation table review. - Contributed to Task 6 application scenario, ethical considerations and future work. - Reviewed the final report, README/user manual and submission folder before final submission.	

Team Agreement	
Team's agreed principles of behaviour, communication, operational processes	The team agreed to communicate regularly through group chat and meetings. Tasks were divided clearly based on the assignment requirements, including model implementation, output generation, evaluation, significance testing, report writing and final checking. Both members agreed to share progress, files and results with each other for checking before submission. The team agreed to be respectful, respond to messages within a reasonable time, inform the other member early if there were any delays, and review the final work together before submission. The team also agreed to use only the assignment specification, provided datasets and tutorial/workshop-based code logic for implementation.
Non-compliance	Minor: Minor non-compliance includes late replies, small delays in completing allocated tasks, formatting issues, or minor mistakes in code/report sections. These were to be managed through direct communication, reminders and task adjustment where required. Major: Major non-compliance includes not completing assigned work, not communicating for an extended period, submitting incorrect or untested work without informing the team, or refusing to contribute fairly. Major issues would be documented and raised with the tutor if they affected fairness or project completion.
Conflict Management	Manage Minor Issues: Minor issues were managed through discussion between the two team members. If a task was delayed or unclear, the team clarified responsibilities, adjusted the workload and set a new deadline. The aim was to resolve issues quickly without affecting the final submission. Manage Major Issues: Major issues would be managed by recording the issue, discussing it with both team members, and attempting to agree on a fair solution. If the issue could not be resolved internally, the team would contact the tutor for guidance and provide evidence of individual contributions if required.
Other team issues or comments	No major team issues were recorded. Both members contributed to the completion of the assignment and reviewed the final submission materials.

User Manual

- Information on used packages or any GenAI support:

  The implementation uses Python 3.x and built-in modules only: os, re, glob, zipfile, string, collections, csv and math. It also uses the provided common-english-words.txt stop-word list and the local Porter2 stemming file in stemming/porter2.py. No additional package installation is required.

  The implementation follows the unit workshop style for XML parsing, bag-of-words document representation, stop-word removal, stemming, BM25 ranking, JM language modelling, pseudo-relevance expansion, AP/P@10/DCG evaluation and paired significance testing. File handling and report-output generation were organised to apply these methods across all 50 datasets.

- Data folder setup:

  The data folder must contain Topics.txt, Doc_Collection and Relevant Judgements. When the main runner is executed, Doc_Collection and Relevant_Judgements are extracted if the corresponding folders are not already present.

- How to execute Python files:

  Open the project folder in PyCharm or another Python IDE and run:
  Python Task3_Run_All_Models.py

- Expected outputs for each Python file:

  The main runner creates Baseline1_R101_Ranking.dat to Baseline1_R150_Ranking.dat, Baseline2_R101_Ranking.dat to Baseline2_R150_Ranking.dat, and ModelC_R101_Ranking.dat to ModelC_R150_Ranking.dat in ModelOutputs. Each ranking file uses the format: Doc_ID Score.

  The EvaluationOutputs folder contains per_topic_evaluation.csv, evaluation_summary.csv, significance_tests.csv, top10_rankings.csv, Appendix_Top10.txt, Validation_Table.txt and model_c_expansion_terms.txt.

- Additional information (if any)

  Each ranking file has two columns document ID and score which are sorted in descending order of score, with no header line, matching the format shown in the specification.
  Current Evaluation Summary:

Model	MAP	Average_P@10	Average_DCG@10
Baseline1	0.491078	0.358000	2.090029
Baseline2	0.458335	0.344000	1.942974
Model_C	0.497682	0.372000	2.149772

Design

Task 1: Design two baseline models: Extend the BM25-based IR model and language models to rank documents in each dataset according to the corresponding topic.

Baseline1:

- Description of the approach for applying a BM25-based IR model to the provided data collection as a baseline.

- Baseline1 ranks the documents of each dataset using the BM25 model defined in Equation 1 of the specification. For each topic, the topic title is used as the query. Every dataset is processed independently. Its documents are parsed into a bag-of-words representation, the document frequency of each term and the average document length are computed, and each document is scored against the title query. Documents are then sorted by descending BM25 score and saved to a ranking file.

- Assumption (if any)

- The topic title (not the description or narrative) is used as the query, as required by the specification.
- Documents and queries receive identical pre-processing (digit removal, punctuation removal, lower-casing, Porter2 stemming, stop-word removal, and removal of terms of two characters or fewer).
- The BM25 constants are fixed by the specification: k1 = 1.2, k2 = 500, b = 0.75.

- Algorithm 1 (including inputs, outputs, and the procedure or method in plain English or pseudocode)

- 1. Read all topics from Topics.txt and loop through R101 to R150.
  2. For topic Ri, select the title as query Qi and select corresponding Dataset.
  3. Parse every XML document in Dataset and represent each document as a term-frequency dictionary with document length.
  4. Calculate document frequency nt for each term and calculate average document length for Datset.
  5. For each document D, calculate BM25(D,Qi) using k1 = 1.2, k2 = 500 and b=0.75.
  6. Sort all document in descending order of BM25 score.
  7. Save the ranking as ModelOutputs/Baseline1_Ri_Ranking.dat.

- Description of the meaning of the parameters N, n_t, f_t and qf_t within the algorithm.

- N – The total number of documents in the dataset.
- n_t - The number of documents in the dataset that contain term t (its document frequency).
- f_t – The number of times term t occurs in the current document D.
- qf_t – The number of times term t occurs in the query (the topic title).

Baseline2:

- Description of the approach for applying the language model model to the provided data collection as a baseline.

- Baseline2 ranks documents using the JM smoothing language model defined in Equation 2, with $\lambda = 0.3$. For each document, the probability of each query term is estimated by mixing the term's probability in the document with its probability in the whole dataset. To avoid numerical underflow from multiplying many small probabilities, the score is computed as the sum of base-10 logarithms of these mixed probabilities. This preserves the ranking and produces the negative log-scores show in the specification's example output.

- Assumptions (if any)

- The topic title is used as the query.
- $\lambda = 0.3$, as specified.
- If a query term occurs in neither the document nor the dataset, a very small probability (1e-12) is substituted to avoid log(0), such a term contributes a large negative value, keeping that document low in the ranking.

- Algorithm 2 (including inputs, outputs, and the procedure or method in plain English or pseudocode)

- 1. Read all topics from Topics.txt and loop through R101 to R150.
  2. For topic Ri, select the title as query Qi and select the dataset.
  3. Parse all XML documents and store document term-frequency dictionaries and document lengths.
  4. Calculate collection term frequencies and total word occurrences in Dataset.
  5. For each document D, calculate the JM-smoothed log likelihood score using lambda = 0.3.
  6. Sort all documents in descending order of JM score.
  7. Save the ranking as ModelOutputs/Baseline2_Ri_Ranking.dat.

- Description of the meaning of the parameters n and c_{qj} within the algorithm.

- n – In the JM model the relevant counts are f_qj,D, the number of times query term q_j occurs in document D, and its collection count c_qj. (The symbol n denotes document frequency in the BM25 model; the JM model does not use n directly but instead relies on these within-document and collection term-frequency counts.)
- Cq_j – The number of times query term q_j occurs in the whole dataset (its collection frequency), used in the collection-probability term c_qj/|Dataset_i|.

Task 2. Design a Custom Model (Model_C)

Design your own ranking model leveraging knowledge from this unit, including learning-to-rank and pseudo relevance techniques, to rank documents in
each dataset.

- Model_C is a hybrid pseudo-relevance feedback model that combines the two baseline scoring methods. It first runs BM25 on the topic title to obtain an initial ranking, then applies pseudo-relevance feedback by assuming the top-ranked documents are relevant and expanding the query with a term taken from them, and finally re-ranks all documents using a weighted combination of normalised BM25 and JM scores. The approach is generic: it uses only the topic title and the documents themselves, so it can be applied to any topic without manual tuning per query.
- Data Structures: A single document is represented by a BowDoc object holding its document ID, a Counter of term frequencies, and its length. The whole dataset is represented by a BowColl object, a dictionary mapping document ID to BowDoc. Document frequency is a dictionary {term: count}, and collection term frequency is a Counter. Scores for a dataset are held in a dictionary {document_id: score}.

- Assumption (if any).

- The top 5 BM25-ranked documents are treated as pseudo-relevant (no human judgements are used at ranking time).
- One expansion term is added, selected by frequency across the pseudo-relevant documents after removing query words, common Reuters/XML tokens, and terms of three characters or fewer.
- BM25 and JM scores are on different scales, so each in min-max normalised to the range 0-1 before they are combined.

- Algorithm 3 (including inputs, outputs, and the procedure or method in plain English or pseudocode).

- 1. For each topic Ri, read the title query Qi and parse the corresponding Dataset.
  2. Calculate document frequency and collection term frequency statistics for Dataset.
  3. Run BM25 using Qi to obtain an initial ranking.
  4. Select the top_docs highest-ranked documents as pseudo-relevant documents.
  5. Count candidate expansion terms in the pseudo-relevant documents and remove original query terms, blocked artefact terms and unsuitable short/non-alphabetic terms.
  6. Select expansion_terms terms and add them to the original query with expansion_weight.
  7. Run BM25 and JM again using the expanded query.
  8. Min-max normalise the expanded BM25 and JM scores.
  9. Calculate the final score as alpha times normalised BM25 plus (1-alpha) times normalised JM.
  10. Sort documents by final score and save ModelOutputs/ModelC_Ri_Ranking.dat.

- Discussion of the differences between *Model_C* and the other two baseline models.

- Query: The baselines use the raw title, Model_C expands the title with one feedback term.
- Feedback: The baselines use none, Model_C uses pseudo-relevance feedback over the top 5 documents.
- Scorers: Each baseline uses a single scorer (BM25 or JM), Model_C fuses both.
- Score scale: The baselines output raw scores, Model_C normalises both score sets to 0-1 before blending so the weighting is meaningful.

Implementation

Task 3. Implement the three models in Python and test them on the provided datasets for their respective topics.

Baseline1:

- Python package or modules (or any open-source software) used:
- Standard library only: math (logarithms), plus the shared loader in Task3_Data_Loader.py which uses re, os, glob, string, collections.Counter and the Porter2 stemmer. Implemented in Task1_Baseline1_BM25.py as bm25_scores().

- Data structures used to represent individual documents and the overall document collection:
- Each document is a BowDoc (Id, Counter of term frequencies, length). The dataset is a BowColl (dictionary of BowDoc by ID). Document frequency is a {term: count} dictionary.

- Screenshots similar to those shown in the Assignment 2 specification:

Example ranking in Baseline1_R135_Ranking.dat:

```
Doc_ID BM25_Score
57876 0.6744874940228573
71894 0.5196381472811176
42848 0.5172924578571716
2738 0.40620267332577525
79698 0.030066821554749767
78998 0.030066821554749767
65311 0.03006577590333607
65501 0.030063613688166558
52721 0.030049043784249163
80891 0.030028313847379407
34733 0.02971569244014226
67433 0.02965851628715475
67275 0.02965851628715475
65465 0.029644352811617418
66118 0.029644352811617418
59518 0.029585217818656463
49229 0.029542582822174028
46122 0.029542582822174028
68756 0.029457156492224544
44221 0.029163304248653862
41556 0.028228417380549337
40131 0.028228417380549337
82689 0.0280563112567677
78305 0.027931110985616882
```

Baseline2:

- Python package or modules (or any open-source software) used:

- Python file: Task1_Baseline2_JM.py uses JM smoothing with lambda = 0.3.
  Modules used: Python built-in modules plus local stemming/porter2.py and common-english-words.txt. No additional external package installation is required.

- Data structures used to represent individual documents and the overall document collection:

- BowDoc stores a document ID, term-frequency dictionary and document length. BowColl stores the set of documents for a dataset as a dictionary. Additional dictionaries/Counters store document frequencies, collection term frequencies and ranking scores.

- Screenshots similar to those shown in the Assignment 2 specification:

```
Doc_ID JM_Score
78305 -7.343891396659082
65465 -7.372657561247056
66118 -7.372657561247056
31784 -7.476367943795756
68756 -7.492650695242742
80891 -7.51191507746273
79698 -7.585546266969745
78998 -7.585546266969745
52721 -7.589256358710838
65311 -7.595775792578905
65501 -7.597396961971727
67433 -7.627770861181642
67275 -7.627770861181642
34733 -7.6633111714308715
86691 -7.675503492819451
7741 -7.770504591276963
66181 -7.778809630670664
82689 -7.7914442969716085
49229 -7.796986956267717
46122 -7.796986956267717
59518 -7.838849099648349
44221 -7.916616095131944
41556 -8.11795200619079
40131 -8.11795200619079
```

Model_C:

- Python package or modules (or any open-source software) used:
  - Python file: Task2_ModelC.py. Uses pseudo-relevance expansion and hybrid BM25/JM re-ranking.
    Modules used: Python built-in modules plus local-stemming/porter2.py and common-english-words.txt. No additional external package installation is required.

- Data structures used to represent individual documents and the overall document collection:
  - BowDoc stores a document ID, term-frequency dictionary and document length. BowColl stores the set of documents for a dataset as a dictionary. Additional dictionaries/Counters store document frequencies, collection term frequencies and ranking scores.

- Screenshots similar to those shown in the Assignment 2 specification:

```
Doc_ID ModelC_Score
65465 0.9650000302914801
66118 0.9650000302914801
31784 0.9109924224914374
66181 0.8533130305888551
65311 0.7296117259562751
65501 0.7283992857611242
57876 0.6780459654026381
79698 0.6410926486067754
78998 0.6410926486067754
78305 0.632684013814416
71894 0.6042914889428148
49229 0.5982775374863908
46122 0.5982775374863908
44221 0.5879002968454736
42848 0.5759803426613128
82689 0.5754040689046866
68756 0.5500100730578218
59518 0.5448937175721982
80891 0.5415865963454706
2738 0.5034965664965977
52721 0.49973258106935436
67433 0.47857321866372643
67275 0.47857321866372643
41556 0.4618460628467021
```

Evaluation, Validation and Significant Test

Task 4. Evaluation and validation strategy.

(a) Summarize evaluation results:

All three models were evaluated on all 50 topics (R101-R150) using Average Precision (and MAP), precision@10 and DCG@10. The per-topic results are given in Tables 1-3. The averages are shown in the final row of each table.

Table 1: The performance of 3 models on average precison (MAP)

Topic	Baseline1	Baseline2	Model_C
R101	0.684892	0.684892	0.530130
R102	0.792497	0.789101	0.810528
R103	0.383498	0.344907	0.438516
R104	0.915328	0.838239	0.929814
R105	0.769689	0.650766	0.701343
R106	0.425245	0.414001	0.421970
R107	0.358974	0.304094	0.300000
R108	0.128472	0.128307	0.179654
R109	0.634641	0.662094	0.660993
R110	0.486667	0.349017	0.608970
R111	0.130754	0.130754	0.177273
R112	1.000000	0.944444	1.000000
R113	0.635349	0.570070	0.710078
R114	0.885000	0.926667	0.826190
R115	0.325397	0.237923	0.293478
R116	0.301108	0.259686	0.265426
R117	0.268687	0.483333	0.521368
R118	0.294444	0.224074	0.261905
R119	0.132129	0.132129	0.209878
R120	0.523499	0.561255	0.460714
R121	0.600784	0.699160	0.755242
R122	0.552790	0.554934	0.575696
R123	0.477273	0.420168	0.410534
R124	0.390517	0.436598	0.312500
R125	0.717751	0.542520	0.720893
R126	0.582473	0.608663	0.735130
R127	0.574603	0.423290	0.543290
R128	0.328084	0.143373	0.133352
R129	0.497738	0.469903	0.715071
R130	0.698413	0.591667	0.700000
R131	0.424575	0.251786	0.159295
R132	0.230713	0.225728	0.242997
R133	1.000000	1.000000	1.000000
R134	0.179038	0.197016	0.207316
R135	0.382114	0.546128	0.712237
R136	0.365443	0.435232	0.477846
R137	1.000000	0.866667	1.000000
R138	0.209509	0.215967	0.303095
R139	0.866667	0.738095	0.791667
R140	0.812766	0.693124	0.666924
R141	0.843620	0.703054	0.795371
R142	0.177137	0.177137	0.202381
R143	0.130277	0.106634	0.118437
R144	0.376098	0.405373	0.496368
R145	0.045437	0.057458	0.047037
R146	0.382018	0.343557	0.343557
R147	0.644298	0.365192	0.377660
R148	0.367578	0.367578	0.367578
R149	0.200481	0.208873	0.231084
R150	0.419444	0.486111	0.433333
MAP	0.491078	0.458335	0.497682

Table 2: The performance of 3 models on precision@10

Topic	Baseline1	Baseline2	Model_C
R101	0.500000	0.500000	0.500000
R102	0.600000	0.600000	0.700000
R103	0.500000	0.400000	0.500000
R104	1.000000	0.700000	0.900000
R105	0.700000	0.700000	0.700000
R106	0.200000	0.200000	0.200000
R107	0.200000	0.200000	0.200000
R108	0.100000	0.100000	0.100000
R109	0.600000	0.700000	0.500000
R110	0.400000	0.400000	0.400000
R111	0.000000	0.000000	0.100000
R112	0.600000	0.600000	0.600000
R113	0.600000	0.600000	0.700000
R114	0.500000	0.500000	0.500000
R115	0.200000	0.200000	0.200000
R116	0.200000	0.100000	0.300000
R117	0.200000	0.200000	0.200000
R118	0.300000	0.200000	0.300000
R119	0.000000	0.000000	0.100000
R120	0.600000	0.500000	0.300000
R121	0.600000	0.700000	0.800000
R122	0.400000	0.400000	0.400000
R123	0.100000	0.100000	0.100000
R124	0.400000	0.400000	0.300000
R125	0.700000	0.700000	0.700000
R126	0.700000	0.600000	0.700000
R127	0.500000	0.400000	0.400000
R128	0.100000	0.100000	0.200000
R129	0.400000	0.600000	0.700000
R130	0.300000	0.300000	0.300000
R131	0.100000	0.100000	0.100000
R132	0.200000	0.200000	0.200000
R133	0.500000	0.500000	0.500000
R134	0.100000	0.100000	0.200000
R135	0.200000	0.400000	0.600000
R136	0.400000	0.400000	0.300000
R137	0.300000	0.300000	0.300000
R138	0.100000	0.100000	0.100000
R139	0.300000	0.200000	0.300000
R140	0.900000	0.900000	0.800000
R141	0.900000	0.600000	0.900000
R142	0.100000	0.100000	0.200000
R143	0.000000	0.000000	0.000000
R144	0.300000	0.400000	0.400000
R145	0.000000	0.000000	0.000000
R146	0.100000	0.100000	0.100000
R147	0.500000	0.400000	0.300000
R148	0.100000	0.100000	0.100000
R149	0.200000	0.200000	0.200000
R150	0.400000	0.400000	0.400000
Average	0.358000	0.344000	0.372000

Table 3: The performance of 3 models on DCG_{10}

Topic	Baseline1	Baseline2	Model_C
R101	3.247425	3.247425	2.747425
R102	3.434024	3.434024	3.605696
R103	2.873990	2.431960	3.035651
R104	5.254495	4.207323	4.823818
R105	4.022788	2.754495	3.552177
R106	1.386853	1.356207	1.430677
R107	1.500000	1.130930	1.130930
R108	0.333333	0.315465	0.500000
R109	3.707323	4.120131	3.286884
R110	1.948459	1.710405	2.689541
R111	0.000000	0.000000	0.301030
R112	3.948459	3.877071	3.948459
R113	3.747425	3.218843	4.195970
R114	3.394940	3.448459	3.243060
R115	1.130930	0.886853	1.333333
R116	1.315465	1.000000	1.058525
R117	0.931960	1.301030	1.386853
R118	1.187883	0.746141	1.058525
R119	0.000000	0.000000	0.301030
R120	3.189455	2.894940	2.130930
R121	3.710405	4.189455	4.439030
R122	2.005005	2.005005	2.173737
R123	1.000000	1.000000	1.000000
R124	2.189541	2.286884	1.689541
R125	4.210405	3.398287	4.207323
R126	2.754495	2.934024	3.808100
R127	2.472702	2.044090	2.076393
R128	1.000000	0.500000	0.634363
R129	2.964263	3.195970	4.195970
R130	1.987137	1.764010	1.930677
R131	1.000000	1.000000	0.356207
R132	1.430677	1.430677	1.630930
R133	3.561606	3.561606	3.561606
R134	0.386853	0.630930	0.931960
R135	0.689541	2.431960	3.948459
R136	2.134363	2.116495	2.061606
R137	2.630930	2.430677	2.630930
R138	1.000000	0.356207	1.000000
R139	2.430677	2.000000	2.333333
R140	4.823818	4.254495	3.754495
R141	4.898287	3.547171	4.953465
R142	0.315465	0.315465	0.720186
R143	0.000000	0.000000	0.000000
R144	1.720186	1.751147	1.948459
R145	0.000000	0.000000	0.000000
R146	1.000000	1.000000	1.000000
R147	2.948459	1.802602	1.987137
R148	0.301030	0.301030	0.301030
R149	0.616495	0.616495	0.634363
R150	1.763924	2.202318	1.818813
Average	2.090029	1.942974	2.149772

Analysis: Model_C achieves the highest score on all three measures MAP 0.498, Precision@10 0.372, and DCG@10 2.150 ahead of both Baseline1 (BM25: MAP 0.491, Precision@10 0.358, DCG@10 2.090) and Baseline2 (JM: MAP 0.458, Precision@10 0.344, DCG@10 1.943). The gain over Baseline2 is larger than the gain over Baseline1 on every measure, which is consistent with the significance tests in Task 5, where Model_C is significantly better than Baseline2 but not significantly different from Baseline1. The improvement is consistent across all three measures, indicating that the pseudo-relevance query expansion and the BM25–JM blend together produce a small but uniform gain in ranking quality rather than helping only one metric.

(b) Parameters and their roles and impacts and the validation process

Model_C has four tuned parameters:

- α (alpha): The weight given to the normalised BM25 score in the final blend. (1-α) is given to the JM score. A smaller α relies more on the JM language model.
- λ (lambda): The JM smoothing weight inside Model_C. It balances the document probability against the collection probability.
- top_docs: The number of top BM25 documents treated as pseudo-relevant for feedback.
- expansion_terms: The number of new terms added to the query from the pseudo-relevant documents.

Validation process: systematic variation was used over the whole collection. Starting from the chosen setting (α = 0.05, λ = 0.1, top_docs = 5, expansion_terms = 1), each parameter was varied in turn (α to 0.10 and 0.20; λ to 0.3; top_docs to 10; expansion_terms to 3) and the three measures were recomputed across all 50 topics. The chosen setting gave the best MAP, Precision@10 and DCG@10, and was therefore selected as the final configuration. Increasing α shifted weight toward BM25 and slightly reduced MAP, increasing the number of expansion terms added noise from the pseudo-relevant documents and did not improve MAP. These results support the selected values: a small α (JM-dominant blend) with a single, high-quality expansion term.

Table 4: Model_C validation and parameter selection results

Configuration	MAP	P@10	DCG@10
Baseline1: BM25 title query	0.491078	0.358000	2.090029
Baseline2: JM title query	0.458335	0.344000	1.942974
Model_C alpha=0.05 lambda=0.1 top_docs=5 terms=1	0.497682	0.372000	2.149772

Task 5. Significance Test

A paired t-test was used to compare Model_C against each baseline across the 50 topics, for each of the three measures. The test was implemented by hand using the math module (the t-statistic is exact, the p-value uses a normal approximation through the complementary error function), because SciPy is not in the approved library list. A p-value below 0.05 is treated as statistically significant.

Conduct statistical tests to determine whether your custom model (Model_C) outperforms the two baseline models.

- Report the results of the t-test

Comparison	Measure	t-statistic	p-value	significant
Model_C vs Baseline 1	MAP	0.420	0.674	No
	P@10	0.926	0.355	No
	DCG@10	0.650	0.516	No
Model_C vs Baseline 2	MAP	3.361	0.0008	Yes
	P@10	2.137	0.0326	Yes
	DCG@10	2.927	0.0034	Yes

- Discuss the results for using the effectiveness measures
- Using a 0.05 significance threshold, Model_C significantly outperforms Baseline2 on Average Precision, Precision@10 and DCG@10. Model_C does not significantly outperform Baseline1 on the three measures. Therefore, Model_C is recommended cautiously: it achieves the highest average results and provides statistically significant improvement over the JM baseline, but not statistically significant improvement over the BM25 baseline.

- Conclusion
- The significance test shows that Model_C provides mixed but useful evidence. Model_C significantly outperforms Baseline2 for all three effectiveness measures, as the p-values for Average Precision, Precision@10 and DCG@10 are all below 0.05. This means the improvement over the JM baseline is statistically significant. However, Model_C does not significantly outperform Baseline1, because the p-values for Model_C vs Baseline1 are greater than 0.05 for all three measures. Therefore, the null hypothesis cannot be rejected for the comparison with the BM25 baseline. Overall, Model_C achieved the highest average MAP, Precision@10 and DCG@10, and it shows statistically significant improvement over Baseline2. However, because the improvement over Baseline1 is not statistically significant, Model_C should be recommended cautiously rather than claimed as significantly better than both baselines.

Application Scenario and future work

Task 6. Describe a potential application scenario. Discuss how your model could be extended to improve performance in the future and consider any relevant ethical issues

(a) Application Scenario and Ethical considerations:

- A realistic application scenario for Model_C is a news archive search system for journalists, researchers or policy analysts. Users often submit short topic-style queries, and the system must retrieve relevant news articles from a large document collection. Model_C is suitable for this scenario because it starts with the original query, uses pseudo relevance feedback from initially high-ranked documents, expands the query using related terms, and then re-ranks documents using both BM25 and JM evidence.
- Ethical considerations:
- Data privacy: News archives may contain personal information about named individuals, access controls and compliance with privacy law are required.
- Bias in the collection: pseudo-relevance feedback assumes the top documents are relevant, so any topical or political bias in the archive can be amplified through query expansion.
- Fairness across user groups: The same ranking may serve users with different information needs unequally if certain topics are under-represented in the collection.
- Transparency and explainability: Because Model_C blends two scores and adds an automatically chosen term, users may not understand why a document was ranked highly.

(b) Approaches for improving ranking performance in the future and mitigation strategies for the identified ethical issues:

- Directions for improving ranking performance:
- Select expansion terms by how well they discriminate relevant from non-relevant documents (for example a relevance-weighting score) rather than by raw frequency and test a small number of expansion terms with stronger noise filtering, to aim for a statistically significant gain over BM25.
- Tune parameters on a separate validation split rather than the whole collection, so the chosen values generalise and are not over-fitted to these 50 topics.
- Explore a learning to rank approach that learns the combination weight α from data instead of fixing it.

- Mitigation strategies for the ethical issues:
- Privacy: Restrict access, log queries responsibly, and remove or mask personal data where it is not needed.
- Bias: Monitor which documents are repeatedly used for feedback and audit the collection for over represented sources, allow feedback to be disabled.
- Fairness: Evaluate performance separately across topic groups and report disparities.
- Transparency: Shows users the expansion term that was added and indicate which documents contributed to the ranking, so the result can be explained.

References

1. IFN647 **Assignment 2 Specification and Requirements document, Queensland University of Technology.**
2. IFN647 **Data Collection file, Queensland University of Technology.**
3. **IFN647 Evaluation Benchmark file, Queensland University of Technology.**
4. **IFN647 workshop/tutorial codes, Queensland University of Technology.**

Appendix 1 (listing the top-10 ranked documents for each dataset for Baseline1):

Topic	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7	Rank 8	Rank 9	Rank 10
R101	46974	46547	62325	6146	61329	22170	61780	22513	82330	39496
R102	58476	73038	26061	57914	78836	76635	12769	12767	25096	82227
R103	14314	81463	27426	27106	54533	59459	83370	20159	26385	14069
R104	11930	11923	11481	6636	49815	7502	3837	12812	11485	3827
R105	86961	86042	15744	33961	64966	51493	49633	48148	80484	80483
R106	16841	73731	49023	51441	7914	16723	55891	35782	68829	6386
R107	51576	77936	71157	79950	59244	67107	67411	86459	31404	41791
R108	82290	5177	5176	2829	10759	83932	50314	10810	22169	80518
R109	16953	26073	61540	4933	16575	64476	67717	78626	34684	23398
R110	38920	72320	41791	42439	85147	38356	78595	16827	28185	25590
R111	18100	18164	18163	19440	28923	18706	75763	18859	13057	22382
R112	70466	59464	78115	78269	83201	85553	58748	53046	5166	81350
R113	17217	17030	13685	25500	4053	2323	7722	2451	79404	13690
R114	18879	38452	60810	26700	25303	72354	28014	21110	38330	78708
R115	50020	58137	76605	61647	48443	52066	5671	38068	79396	43604
R116	22679	20188	77250	79559	85143	73431	76422	79585	74010	79864
R117	54844	9544	71102	67204	37425	79525	81694	65692	48566	60811
R118	61677	80908	64892	20763	28903	43173	2906	84529	47129	50269
R119	41275	51082	51080	34033	65942	68177	65019	51488	71547	82567
R120	39184	48981	62781	43393	64800	4527	75135	12555	24062	78091
R121	17004	15074	13294	7320	13309	49980	3885	79604	14772	7289
R122	38244	74775	77971	84331	13984	79418	73186	76564	72052	77921
R123	37440	26681	78018	55119	13485	29079	20315	30089	47025	70379
R124	21511	19285	25328	74036	65604	4528	9981	7225	53830	30653
R125	77907	82519	77381	82085	80922	25833	76899	82106	27974	9011
R126	15228	45159	22250	78479	38191	14933	59083	60588	67522	76216
R127	67104	10706	40234	75738	59095	60141	40292	70010	26026	26144
R128	3740	66961	64070	64033	70233	61170	24214	75201	66923	27215
R129	46494	72826	72899	76669	16900	20126	55197	75465	83946	44081
R130	15776	70556	18235	69863	60528	75483	46458	60483	4063	47530
R131	26568	59502	58957	60118	16170	62536	31504	80756	77349	31515
R132	9843	7486	85553	48981	67115	79565	68370	66981	39184	68780
R133	74695	35853	54844	74681	84398	71102	9544	32662	6939	82290
R134	59659	58438	84627	75562	26253	73397	14567	21172	46142	69112
R135	57876	71894	42848	2738	79698	78998	65311	65501	52721	80891
R136	79605	35104	41487	43962	19488	8768	42907	46422	48692	15404
R137	66879	61252	85322	39136	81418	80107	27297	78455	80039	46874
R138	77971	2552	20899	37732	49872	50734	40341	65378	56318	22407
R139	71102	67204	50076	41231	62586	58507	54844	9544	79525	81694
R140	45396	25063	58343	22321	6857	44400	43139	11825	18393	18151
R141	65284	60155	62292	60221	58503	62212	75561	55848	52215	62213
R142	83869	26011	54126	68798	53188	54675	34410	7397	28354	54716
R143	72134	64896	23698	74605	3666	11852	75564	11744	63379	21265
R144	43899	27525	13135	7359	47896	69511	74200	36318	19626	12579
R145	17654	14501	4300	54291	11106	28025	6237	32472	67785	41461
R146	12035	23687	72072	49904	58000	38821	24287	70836	12196	54065
R147	56691	27441	64153	34675	73146	27911	59813	76089	76092	75754
R148	84829	58009	9133	8622	24044	31619	51453	51260	52489	60293
R149	41538	14196	39661	39926	32400	25529	62392	3936	17390	17360
R150	19409	64482	53361	19411	19420	19885	3148	50614	54538	53269

Appendix 2 (listing the top-10 ranked documents for each dataset for Baseline2):

Topic	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7	Rank 8	Rank 9	Rank 10
R101	46974	46547	62325	61329	6146	22170	61780	22513	82330	39496
R102	78836	58476	57914	26061	73038	76635	12769	12767	25096	24515
R103	14314	81463	27106	27426	14069	26385	54533	20159	26386	59459
R104	11930	3837	11923	49815	11960	25205	11481	65394	9591	19224
R105	64966	33961	15744	65210	51493	86961	86042	3008	49633	48148
R106	16841	49023	73731	51441	55891	35782	16723	7914	8085	68829
R107	71157	51576	79950	77936	59244	67107	67411	67673	31404	37330
R108	82290	10759	2829	50314	5177	5176	54765	83932	10810	80518
R109	26073	16953	64476	16575	67717	61540	24340	23398	78626	34684
R110	38920	71167	38356	78595	42439	72320	16827	85147	41791	30597
R111	28923	18706	75763	13057	22382	18100	18164	18163	52020	19440
R112	59464	78269	85553	70466	78115	5166	58748	53046	83201	68780
R113	37816	13685	25500	17217	17030	7722	13690	4053	2323	40234
R114	18879	38452	60810	72354	21110	25303	26700	24474	28014	37685
R115	50020	58137	48443	61647	52066	76605	43604	38068	46714	5671
R116	20188	22679	71542	71543	71555	85143	77250	74017	82616	79559
R117	71102	54844	9544	67204	79525	65692	81694	48566	37425	60811
R118	80908	64892	61677	28903	20763	35104	47129	2906	43173	84529
R119	51082	51080	41275	34033	65942	68177	51488	71547	82567	38614
R120	48981	39184	62781	43393	4527	78091	64800	75135	78092	21067
R121	17004	15074	13294	15415	7320	13309	15410	49980	14556	3885
R122	38244	84331	77971	74775	13984	79418	73186	76564	72052	77921
R123	37440	55119	29079	13485	20315	26681	78018	54794	30089	36083
R124	25328	9981	21511	74036	19285	53830	7225	65633	4528	2345
R125	76899	77907	82519	25833	77381	27974	71913	80922	82085	82106
R126	45159	38191	15228	14933	67522	60588	9384	78479	22250	59083
R127	40234	67104	10706	59095	60141	40292	75738	56741	70010	26026
R128	64033	27215	64070	3740	24214	66961	61170	70233	75201	66923
R129	76669	46494	72826	72899	35148	44081	41297	75465	83428	21791
R130	15776	60528	70556	69863	18235	75483	60483	46458	4063	73550
R131	16170	26568	60118	59502	58957	62536	77349	80756	31504	13484
R132	9843	7486	85553	48981	67115	68370	68780	39184	78577	79565
R133	74695	35853	54844	84398	74681	6939	82290	71102	35755	32662
R134	59659	58438	73397	84627	75562	21172	69112	60928	26253	46142
R135	78305	65465	66118	31784	68756	80891	79698	78998	52721	65311
R136	35104	79605	41487	43962	19488	42907	8768	83238	46422	58438
R137	66879	85322	81418	80107	61252	46874	80039	4650	74200	78455
R138	20899	49872	77971	37732	56318	50734	2552	40341	65378	47714
R139	71102	67204	54844	9544	50076	79525	65692	41231	81694	48566
R140	6857	45396	22321	58343	43139	44400	25063	11825	18393	18151
R141	65284	60155	51691	62292	60221	75561	59351	9954	39864	39304
R142	83869	26011	54126	68798	34410	53188	54675	7397	28354	54716
R143	23698	72134	74605	64896	3666	75564	11852	44232	11744	63379
R144	43899	13135	36318	19626	69511	7359	27525	35258	26257	47896
R145	17654	14501	32232	4300	11106	32472	6237	51917	28025	67785
R146	23687	12035	72072	58000	49904	38821	70836	24287	54065	12196
R147	56691	82440	27441	64153	77181	44101	27911	29390	34675	59813
R148	84829	9133	8622	58009	24044	31619	51260	51453	52489	60293
R149	25529	41538	14196	39661	32400	62392	39926	3936	30390	17390
R150	19409	53361	19411	19420	64482	19885	3148	26648	54538	39907

Appendix 3 (listing the top-10 ranked documents for each dataset for Model_C):

Topic	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Rank 6	Rank 7	Rank 8	Rank 9	Rank 10
R101	61329	46974	46547	62325	6146	22170	61780	22513	82330	39496
R102	58476	26061	78836	76635	24515	73038	25096	57914	12769	4358
R103	27426	27106	54533	26385	14314	81463	63966	83370	59459	26386
R104	3837	11930	11923	49815	11960	12812	11481	25205	11485	7502
R105	64966	15744	86961	86042	51493	65210	2493	2494	33961	80484
R106	16841	73731	49023	35782	16723	76556	55891	31281	32963	86068
R107	71157	59244	79950	77936	78164	51576	37330	66686	17930	71159
R108	2829	5177	5176	10810	82290	10759	50314	54765	80518	83932
R109	16953	26073	16575	64476	67717	4933	65289	23398	15776	25832
R110	38356	42439	71167	72320	78595	38920	41791	85147	16827	27404
R111	75763	18100	18164	18163	19440	18859	22382	47049	18798	45422
R112	59464	78115	70466	78269	85553	83201	53046	58748	81350	5166
R113	17217	17030	13685	25500	13690	43262	37816	79404	2451	24251
R114	38452	18879	72354	25303	26700	21110	60810	28014	24474	78708
R115	58137	76605	46714	43604	50020	79396	48443	61647	52066	38068
R116	20188	77250	79585	32034	73431	22679	70303	3541	81084	71542
R117	71102	9544	67204	54844	81694	60811	79525	65692	48566	37425
R118	80908	28903	64892	61677	35104	50269	43173	2906	20763	28826
R119	51082	51080	34033	65942	41275	68177	51488	62324	65019	67199
R120	48981	39184	43393	62781	78091	71406	69629	78092	69633	45819
R121	15074	13294	17004	7320	13309	15415	15410	14556	49980	14772
R122	38244	13984	74775	77971	73186	76564	72052	84331	77921	79418
R123	37440	55119	20315	26681	13485	5075	78018	47025	36083	7964
R124	21511	19285	61262	13147	14377	69236	9981	74036	78418	75278
R125	77907	77381	82106	82519	25833	27974	80922	82085	76899	73371
R126	38191	14933	15228	45159	67522	59083	71223	6940	78479	86111
R127	67104	10706	59095	60141	40234	75738	26026	26144	56741	62401
R128	64033	64070	24214	66961	70233	61170	75201	3740	27215	37136
R129	46494	72826	72899	75465	75593	44081	76669	75590	83336	78782
R130	15776	70556	69863	18235	46458	60528	75483	60483	47530	4063
R131	16170	60118	58957	59502	62536	77349	26568	39433	80756	63056
R132	9843	7486	67115	85553	48981	79565	68784	68777	66981	68140
R133	74695	54844	84398	74681	35853	82290	35755	62187	6939	74652
R134	59659	58438	73397	75562	69112	3169	21172	26253	60928	74790
R135	65465	66118	31784	66181	65311	65501	57876	79698	78998	78305
R136	35104	41487	43962	79605	46422	19488	8768	83238	48692	76750
R137	61252	66879	85322	81418	80107	46874	80039	4650	74200	78455
R138	2552	65378	56248	56045	20899	49872	77971	37732	59382	59561
R139	71102	67204	50076	41231	21113	54844	9544	62586	79525	65692
R140	58343	6857	44400	13065	25063	45396	18393	18151	22321	43139
R141	65284	60155	60221	62292	39304	58503	62212	55848	52215	85236
R142	83869	26011	54126	68798	53188	28354	54675	45201	34410	41614
R143	74605	75564	64896	63379	72134	23698	3666	11852	44232	11744
R144	43899	47896	27525	69511	27535	35258	71648	74200	13135	24598
R145	17654	14501	8169	17317	35545	54291	18001	45735	41447	35017
R146	49904	12035	23687	72072	58000	38821	70836	24287	54065	12196
R147	56691	27441	64153	44101	70409	81325	34675	41481	77124	27921
R148	51260	52489	31619	58009	9133	8622	24044	84829	51453	60293
R149	39661	14196	41538	32400	39926	25529	72985	17390	62392	17360
R150	19409	19411	19420	53361	64482	19885	3148	23361	50614	54538